DISTRICT EMERGENCY OPERATION CENTER | UDMA RAPID INTELLIGENCE

NATIONAL DISASTER MANAGEMENT FRAMEWORK - SATELLITE SITREP

TARGET SECTOR: Rio de Janeiro · OSCD
REPORT DATE: 10 Oct 2026, 04:47 UTC
SENSOR / GSD: Sentinel-2 MSI · 10 m
OPERATION: Encroachment / planning audit

TOTAL IMPACT AREA	CRITICAL SITES (P1)	HIGH SEVERITY (P2)	TOTAL DETECTIONS
46,813 m²	0 Zones	3 Zones	7 Polygons

EXECUTIVE SITUATION SUMMARY:

Automated change-intelligence detected 7 change polygons totaling 46,813 m² across the surveyed bounds; 0 critical and 3 high-priority sites are queued for ground verification. Both images were fully observable (no cloud, shadow or no data). Detector: Track B · Sentinel-2 spectral network, F1 22% on OSCD, leave-one-city-out. High-priority footprints are available as draft SACHET-compatible CAP records.

PRIORITY TARGET AUDIT TABLE (TOP DETECTIONS):

ID	AREA (m²)	SEVERITY	CONFIDENCE	CLASSIFICATION	CENTROID (LAT, LON)
CHG_0006	8,237.0	High	98.2%	Water-body Encroachment	-22.9757, -43.4000
CHG_0003	6,931.6	High	98.7%	Vegetation Cleared / Possible Co	-22.9672, -43.3931
CHG_0005	6,328.6	High	98.6%	Vegetation Cleared / Possible Co	-22.9724, -43.3900
CHG_0001	6,228.6	Medium	98.4%	Vegetation Cleared / Possible Co	-22.9639, -43.3775
CHG_0002	6,228.6	Medium	96.1%	Vegetation Cleared / Possible Co	-22.9643, -43.3684
CHG_0004	7,936.3	Medium	96.3%	Ground Disturbance	-22.9685, -43.3652

DATA LIMITATIONS:

- Image comparability 58/100 (low); biggest factor: 169 days apart in the seasonal cycle. Detection threshold raised from 0.915 to 0.9226 outside protected zones.
- Change dated only to the window between the two images: no yearly series stored for this area.
- OpenStreetMap exposure and buffer checks are screening only, not legal boundaries; verify against revenue, FTL and Development Plan records before any notice.

RECOMMENDED IMMEDIATE DIRECTIVES:

1. Send Priority 1-2 sites for field verification in the case workflow.
2. Cross-check flagged sites against Development Plan, CRZ maps and building permissions.
3. Treat 'possible encroachment' as a screening flag, never as a legal finding.
4. Act only on field-confirmed cases; no demolition or eviction on satellite evidence alone.

Prepared By: T.R.A.C.E. Automated Remote Sensing System

Authorised Signatory / District Magistrate / Municipal Commissioner

Signature: _____ Date: _____